Srinath Muppala

+1 7208133485 | srinath.muppala@gmail.com | Boulder, CO (Open to relocation)
LinkedIn - srinath-muppala | GitHub - srinath-19

EDUCATION

University of Colorado Boulder Aug 2024 – May 2026
Masters in Computer Science - Artificial Intelligence GPA - 3.84/4.0

SRM Institute of Science and Technology Sept 2020 – May 2024
Bachelors in Computer Science and Engineering GPA - 3.98/4.0

EXPERIENCE

Inovonics, Boulder, CO - *AI Engineer – Capstone - Algorithms for vectorizing 2D floor plans* Sept 2025 – Present
• Developed a full-stack AI-powered floor plan vectorization platform using **YOLOv12-OBB** for detecting structure with a **FastAPI** backend, **Celery workers**, **Redis task queue**, PostgreSQL database and Next.js frontend deployed on GCP.
• Overcame small-object detection limitations in high-resolution engineering schematics by implementing **tiled sliding-window inference** with overlap-based deduplication for processing high-resolution images and a **9-stage OpenCV image preprocessing pipeline** to clean and normalize noisy architectural drawings before ML inference.
• Designed **Shapely-based room boundary extraction** (wall polygon union, Minkowski gap closing, free-space set-difference) and optimized receiver placement as set-cover using **OR-Tools CP-SAT** with 180-ray coverage analysis.

MPD Industries PVT. LTD., Indore - *Machine Learning Engineer - Internship* May 2025 – Jul 2025
• Prototyped **demand forecasting** (SARIMAX) and **price volume models** (XGBoost) on 500k sales/inventory rows to create SKU and client-level price bands. Early back tests showed 11% MAPE improvement vs. baseline.
• Built a **PuLP profit LP** (15% margin, inventory caps, client rules). Offline simulations reduced sub-margin quotes 12%.
• Ran exploratory **K-Means vendor segmentation** (volume, elasticity, payment terms) to outline tiered pricing options. Scenario analysis suggested a small 1% lift in price realization on select SKUs.

SRM Institute of Science and Technology, Chennai - *Undergraduate Researcher* Aug 2023 – May 2024
• Developed and deployed a real-time **tomato-harvest detection** pipeline on AWS, validating 97% accuracy on a held-out set and delivered a working demo with concise documentation to enable agricultural teams to monitor harvests live.
Publication - 2024 3rd International Conference on Applied Artificial Intelligence and Computing - June 2024
Identification of Harvestable Tomato using Yolov8 **(IEEE Xplore)** (Scopus Indexed) *[Link]* Scopus ID: 59222142100

Aaruush, SRM, Chennai - *Software Developer* Jun 2022 – May 2023
• Designed and deployed a scalable **event management platform** (schedules, activities, ticketing) used by 10k users. Collaborated with peers using Git, using HTML/CSS/JavaScript, Node.js/MongoDB, and AWS/Terraform.
• Maintained 99.9% uptime with **CloudWatch** and Uptime Robot (12 checks/hr, 30 min/month downtime) and validated scale to 5k concurrent users via k6 (15-min soak) p95 380 ms, less than 0.5% errors on **AWS ALB** and CloudFront.

PROJECTS

DealHunter AI - Autonomous Multi-Agent Deal Discovery Engine
Tech Stack – Python, OpenAI API (function calling), LiteLLM, ChromaDB, Modal (severless GPU), Pydantic, Gradio
• Built an **autonomous planning agent** using OpenAI function-calling that independently decides when to scan, price, and notify, running in a self-directed tool-use loop with **zero human intervention**.
• Designed a **5-agent collaborative framework** (Scanner, Frontier, Specialist, Ensemble, Messaging) with a central orchestrator coordinating autonomous runs. Ensemble - **RAG**, **Fine-tuned LLaMA on Modal (severless GPU).**
• ScannerAgent scrapes live RSS feeds, deduplicates against persistent memory, and uses **structured outputs** for validated deal and MessagingAgent uses OpenAI to craft human-friendly push notifications delivered via **Pushover API**.

PricePilot - LLM Fine-Tuning and High-Throughput Serving for Price Prediction
Tech Stack – OpenAI API (batch API, Fine-Tunning API), Hugging Face, QLoRA, bitsandbytes, LiteLLM, vLLM, GCP VM
• Built an end-to-end LLM pricing pipeline, curating an **820K-item supervised dataset** preprocessed via **OpenAI batch processing** (1000-item batches) to produce structured training signals and uploaded splits to Hugging Face Hub.
• Fine-tuned Meta LLaMA 3.2 3B using **QLoRA (4-bit NF4 quantization)** and configured **PEFT/LoRA** targeting attention and MLP projection layers. Trained on A100 GPU using **HF TRL's SFTTrainer** with **bitsandbytes** .
• Tracked training experiments using **Weights & Biases** with periodic model checkpointing to Hugging Face Hub.
• Deployed **vLLM** on a **GCP VM** to serve LLaMA 3.2-3B as base model with **3 LoRA adapters** (request-level routing), sustaining 18 req/s at p95 0.9s for short prompts (200 input tokens) with 55 output tok/s throughput.

GraphFusion RAG - Hybrid graph plus vector retrieval engine
Tech Stack – Python, OpenAI API, LiteLLM, Pydantic (structured outputs), Embeddings, ChromaDB, Pytest, Gradio
• Built a Hybrid **GraphRAG** platform: embedded internal Markdown docs with **OpenAI text-embedding-3-large**, indexed in **ChromaDB**, and combined **vector search with knowledge-graph traversal** to enable multi-hop Q&A, implemented hybrid retrieval with query rewriting, dual semantic retrieval, and **LLM reranking** (top-k candidate selection).
• Developed evaluation framework using **MRR**, nDCG, keyword coverage, and **LLM-as-a-judge** metrics to benchmark and improve answer accuracy and unit tests for graph traversal logic, deduplication, hop limits, and edge-case handling.

SKILLS

Programming - Python, C/C++, SQL, Bash, HTML, CSS, JavaScript/TypeScript.
Technologies &Tools - Git, PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, pandas, Next.js, React, FastAPI, Flask, Redis, Django, SQL/NoSQL databases, Docker, Terraform, Kubernetes, AWS, Google Cloud, Machine Learning, Deep Learning, Computer Vision, NLP, LLM, VectorDB, RAG, Agentic AI, LiteLLM, Pydantic, QLoRA, LangChain.
Certifications - AWS Cloud Operations, Oracle Database, Coursera, Hacker rank *[All certificates' links].*